Assignment #6: Weather and Climate Trend Analysis for Row Crops

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Assignment Overview

Field	Value
Assignment Number	6
Title	Weather and Climate Trend Analysis for Row Crops
Grade Value	8 points
Due Date	March 19, 2026, 11:59 PM PT

Objectives

After completing this assignment, you will be able to:

- Analyze historical weather data for specific row crop field locations.
- Use Python to identify **seasonal patterns** and **climatic anomalies** (e.g., extreme heat or drought).
- Create professional **time-series visualizations** that track weather variables over a growing season.
- Understand how weather trends influence operational decisions like planting dates and irrigation needs.

Instructions

Step 1: Create Your Feature Branch

Open your terminal in VS Code and switch to a new branch for your weather analysis:

```
git checkout -b feature/assignment-06-weather
```

Step 2: Set Up Your Weather Notebook

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1. **Create a new notebook:** notebooks/06_weather_analysis.ipynb.
2. **Load your weather data:** Use the weather dataset (NASA POWER or NOAA) included in your agri-toolkit package.
3. **Data Preparation:** Ensure your date column is in datetime format and filter the data to cover the most recent growing season for your fields.

Step 3: Create a new notebook: notebooks/06_weather_analysis.ipynb.

1. **Load your weather data:** Use the weather dataset (NASA POWER or NOAA) included in your agri-toolkit package.
2. **Data Preparation:** Ensure your date column is in datetime format and filter the data to cover the most recent growing season for your fields.

3. **Task:** Create a time-series visualization showing daily temperature (High/Low) and precipitation for your field locations.
4. **Task:** Calculate a rolling average (e.g., 7-day or 30-day) to smooth out the data and highlight the broader seasonal trend.

Step 4: Identify Climate Anomalies

1. **Task:** Compare this year's weather data against historical averages (if available in your dataset) or identify an "anomaly" such as a late-season frost or a prolonged dry spell.
2. **Task:** Create a visualization specifically highlighting this anomaly (e.g., a bar chart showing rainfall deficit compared to the monthly norm).

Step 5: Design Your Dashboard Weather Plots

1. **Task:** Design **1-2 weather/climate plots** optimized for your final dashboard. These should focus on the most critical weather factors for row crops (Cumulative Precipitation or Heat Units/GDD).
2. **Task:** Save the final plots in output/dashboard_assets/weather_trends.png.

Step 6: Update Your Project Tracker

Open docs/project/assignment-01-tracker.md.

- Document the most significant weather trend you observed.
- Note if any weather stations were missing data and how you handled those gaps (e.g., interpolation or using the nearest station).

Grading Criteria (8 Points Total)

Criterion	Points	Description
Time-Series Analysis	3	Created a clear visualization of seasonal weather patterns (Temp/Precip).

Anomaly Detection	2	Identified and visualized at least one significant climate trend or anomaly.
Dashboard Design	2	Designed 1-2 polished weather plots ready for the final dashboard.
Technical Execution	1	Correct use of Pandas for time-series manipulation and date handling.